

GridBalance: Complete Technical Research Dossier

Exhaustive Repository Audit, Data Quality, ML Experiments, Robustness & Production Deployment

Synthetic Data Provenance Audit

1. Where data.csv came from

data.csv was programmatically generated by the script generate_sample_data.py. The data loader (data_loader.py) is designed to automatically trigger this generator as a fallback if data.csv does not exist in the root folder.

2. Real SGCC Dataset Presence

The real SGCC dataset (which contains ~42,000 customers) is **not present** anywhere in the local repository. A search for .csv or compressed archives yielded only the 1,500-row synthetic data.csv and its generated downstream artifacts.

3. Synthetic Generator Logic (generate_sample_data.py)

- **Normal Customers:** Generated using a baseline uniform random value (5 to 30) added to a sinusoidal seasonal base curve (simulating summer peaks) with 15% random normally-distributed Gaussian noise.
- **Theft Customers:** Theft cases are created by picking normal load profiles and deterministically injecting one of four extreme "theft" mathematical transformations:
 - zeroing: Hardcodes all daily consumption values to 0.0 after a random start day.
 - scaling: Multiplies consumption by a huge reduction factor (e.g. down to 10%-35% of original).
 - partial_bypass: Forces a 60% probability of dropping any given day's consumption to exactly 0.0.
 - intermittent: Forces 40% of the customer's days to exactly 5% of their original value.
- **FLAG Assignment:** FLAG is explicitly assigned 1 for these synthetically tampered rows.
- **Noise:** While base loads have some Gaussian noise, the theft injections (0.0 replacements, extreme scaling) are deterministic structural shocks that bypass the noise completely.
- **Class Distribution:** The theft_ratio was manually hardcoded to 9% to mimic the genuine SGCC dataset class imbalance.

4. Why Engineered Features perfectly predict FLAG

The feature engineering pipeline extracts attributes like zero_ratio, peak_to_average_ratio, min, cv, and longest_zero_streak.

Because the synthetic generator deterministically forces values to 0.0 or applies severe mathematical scalings to the theft class, the theft rows mathematically guarantee extreme outlier values for these engineered features compared to the normal class. Any ML algorithm will find these deterministic separations immediately.